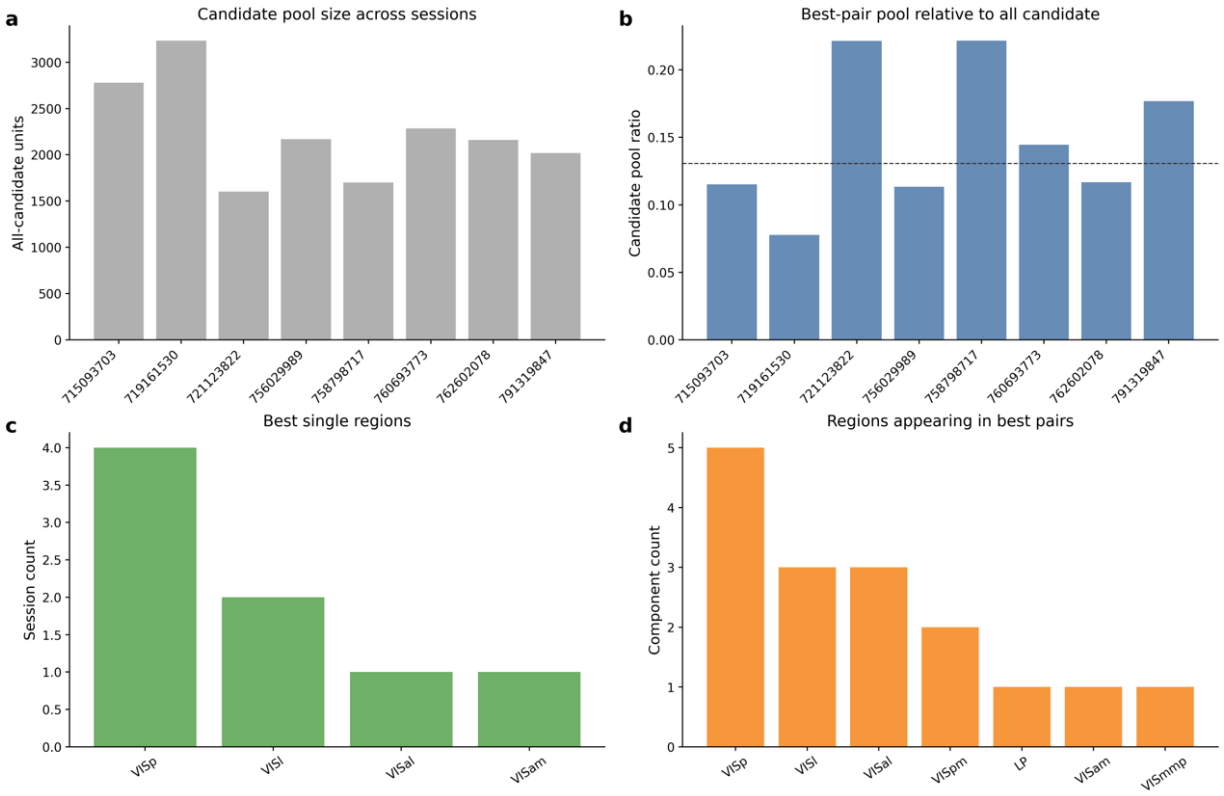
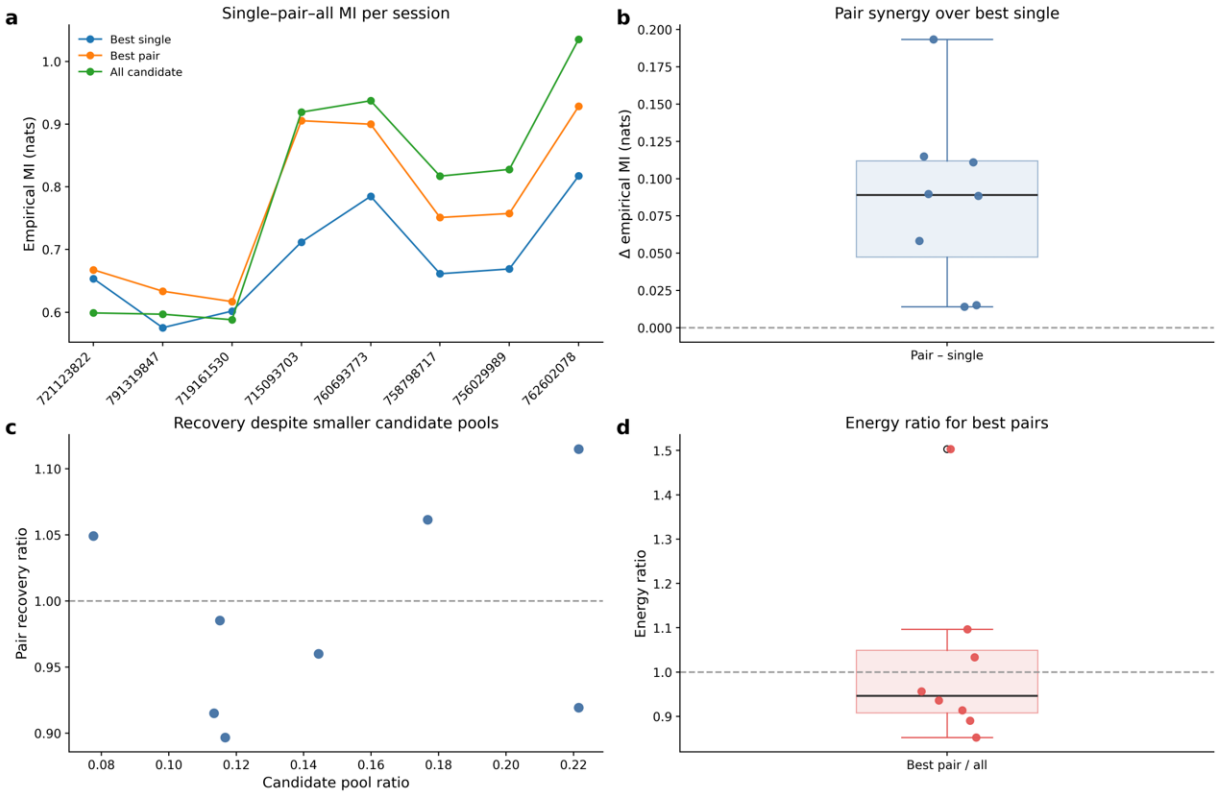
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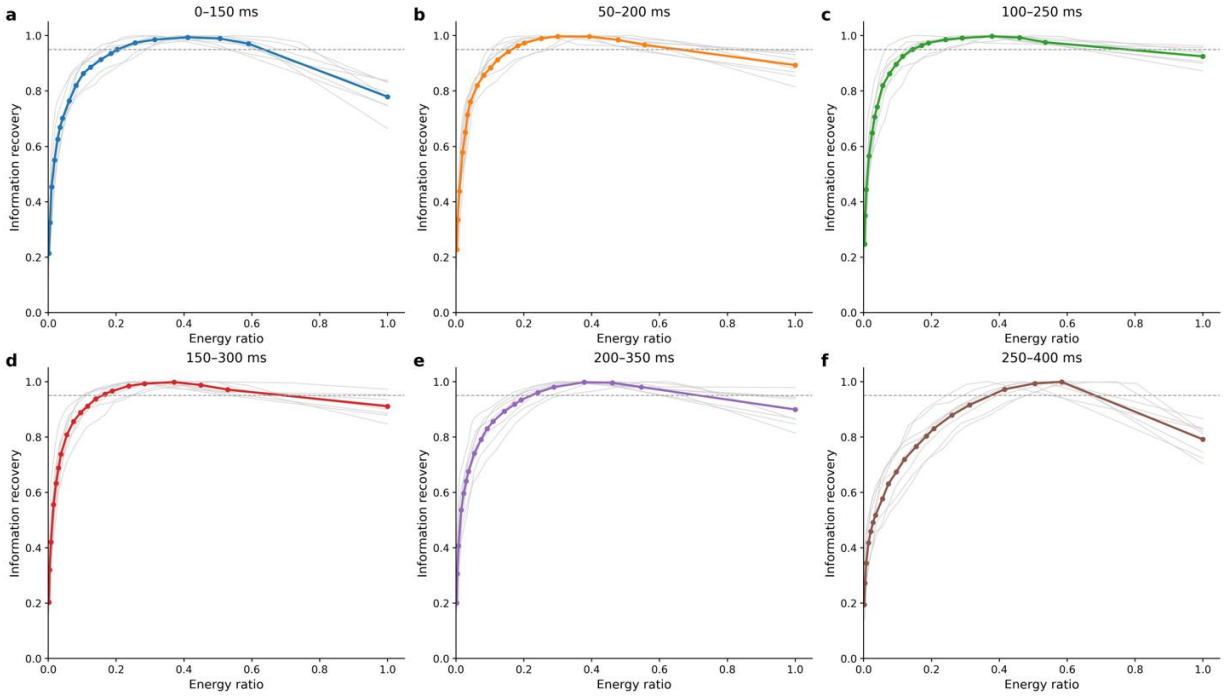
Repository Fig. 1 | Allen session inventory and selected region composition

a, Full candidate-pool size. Number of candidate neurons in each of the eight Allen sessions, illustrating variation in recording scale and candidate-pool size across sessions. **b, Candidate-pool fraction of the best region pair.** Fraction of all candidate neurons contained in the best region pair for each session. The dashed line indicates the cross-session median. Best region pairs typically comprised only a small fraction of the full candidate pool, indicating that regional information recovery did not arise from uniform contributions across all candidate regions. **c, Distribution of best single regions.** Visual regions selected as the best single region across the eight sessions. The best region was not restricted to a single anatomical area but varied across sessions within the visual system. **d, Frequency of regions contributing to the best pair.** Number of times each visual region appeared in a best region pair. Multiple visual areas contributed across sessions, suggesting that structured regional organization reflects session-specific combinations rather than a single fixed region.



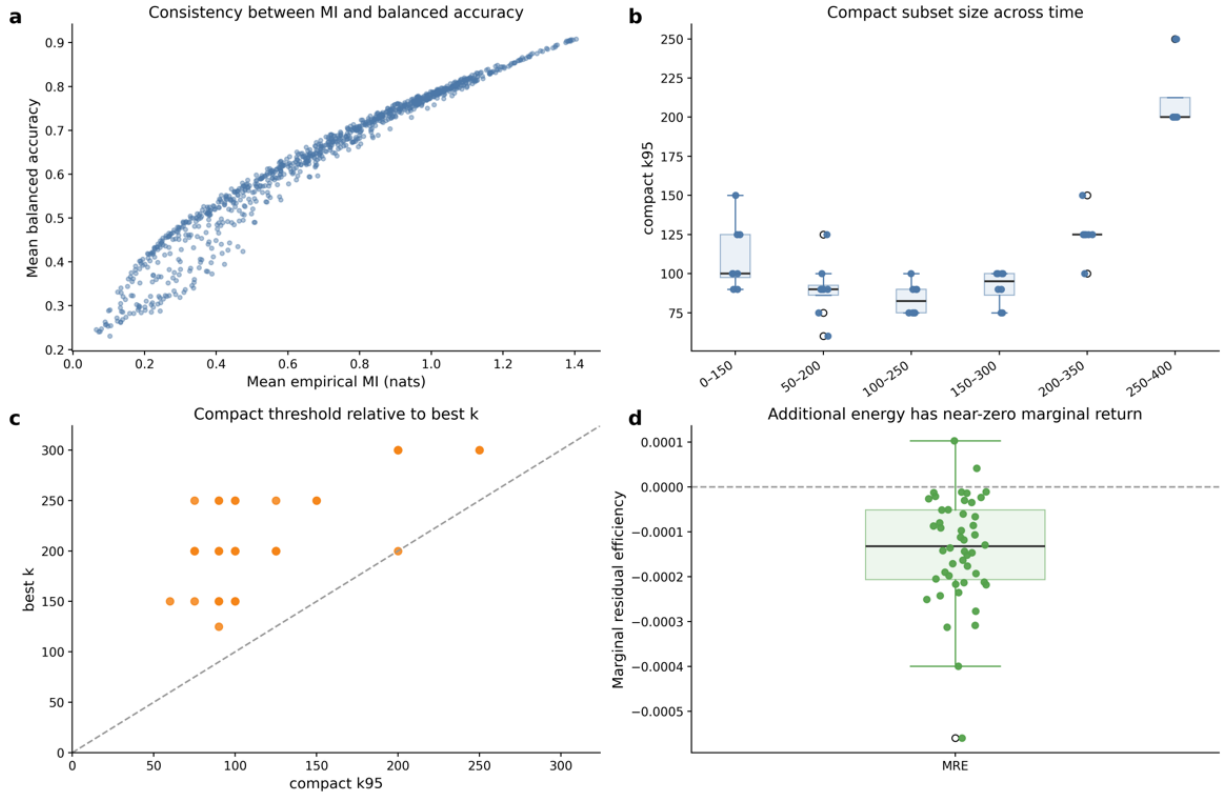
Repository Fig. 2 | Allen single–pair–all complete region-pair results

a, Empirical mutual information across the single–pair–all hierarchy. For each session, empirical mutual information is shown for the best single region, best region pair, and full candidate-region pool. The best pair exceeded the best single region in most sessions and approached or exceeded the full candidate pool. **b, Information gain of the best pair over the best single region.** Each point represents the pair-minus-single empirical-MI difference for one session. Positive values in most sessions indicate additional information contributed by combining complementary regions. **c, Candidate-pool fraction versus information recovery.** The x-axis shows the fraction of the full candidate pool contained in the best pair, and the y-axis shows information recovery relative to the full candidate-region pool. Small regional candidate pools frequently recovered information close to the full population, consistent with structured concentration of visual information. **d, Spike-count energy ratio of the best pair relative to the full candidate pool.** Each point represents one session. These regional analyses primarily establish structured regional organization and information recovery; stricter quantification of spiking-resource savings is provided by the subsequent single-unit top-k frontier analysis.



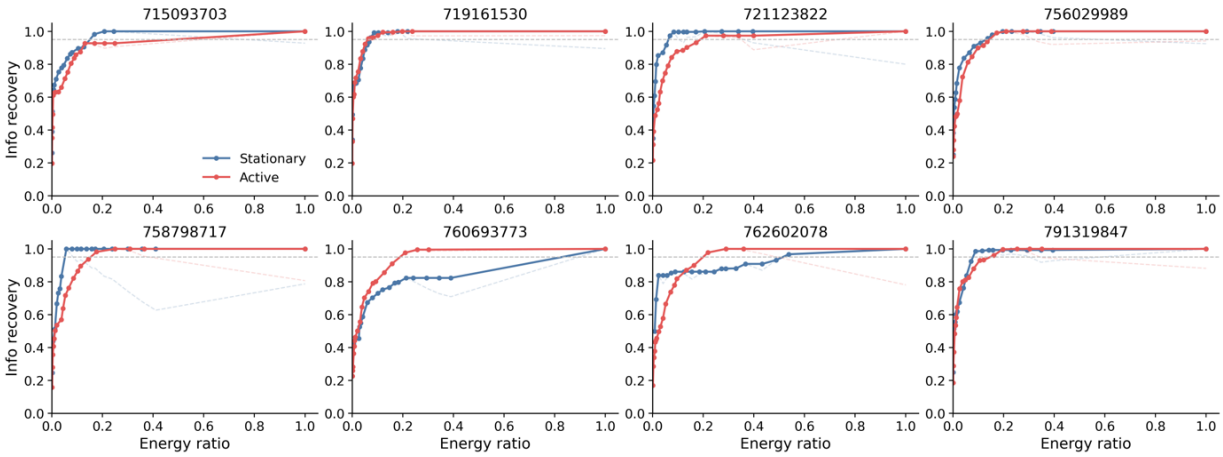
Repository Fig. 3 | Allen top-k frontier curves across all session-windows

a–f, Top-k energy–information frontiers across six overlapping stimulus-response windows. Thin gray lines indicate session-specific top-k frontiers and thick colored lines summarize the cross-session trend. The x-axis shows spike-count energy relative to the all-units configuration, and the y-axis shows information recovery relative to the maximum task information within each session-window. The six windows were 0–150, 50–200, 100–250, 150–300, 200–350, and 250–400 ms. Across most windows, information recovery increased rapidly at low energy ratios and approached a plateau, indicating substantial concentration of visual task information within relatively small top-k populations. Frontier saturation was slower in the latest 250–400 ms window, consistent with increased late-stage recovery cost or greater non-task-specific activity.



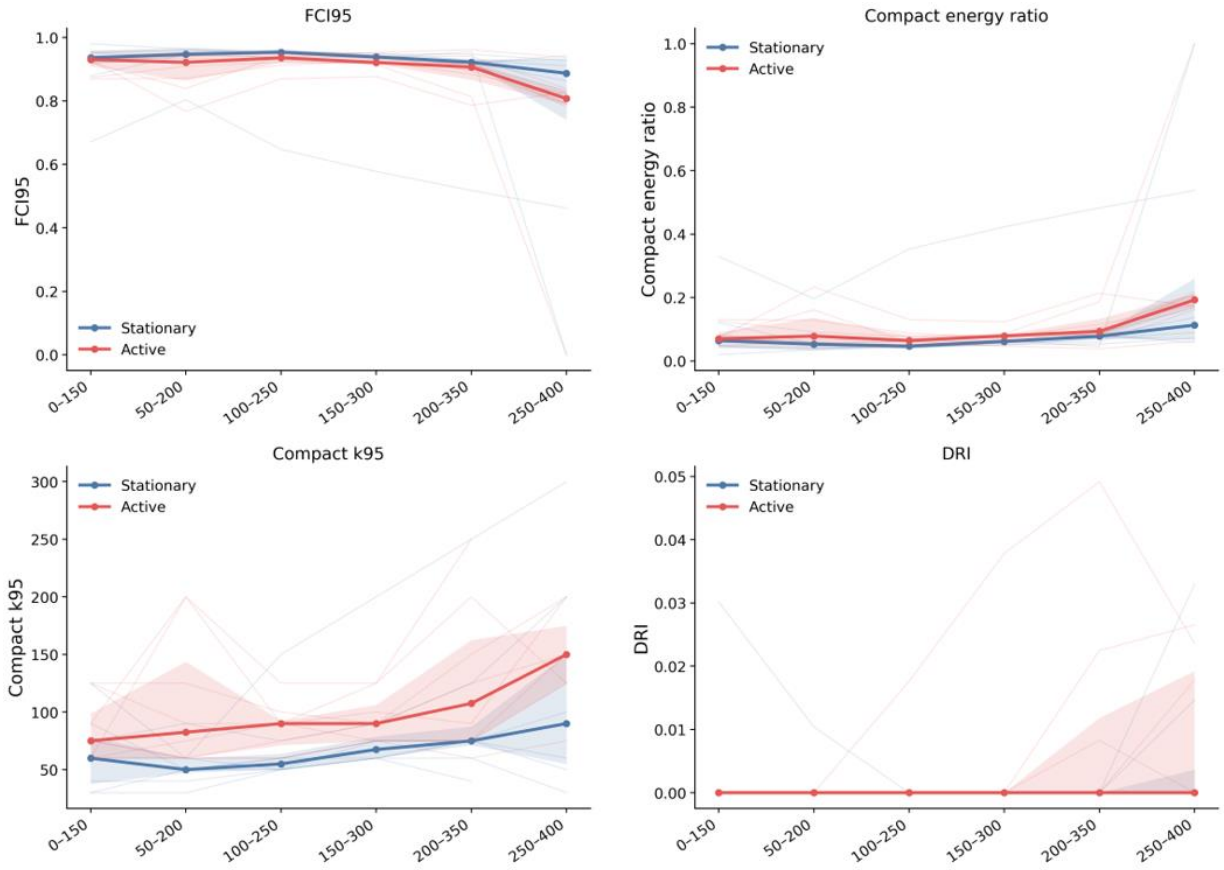
Repository Fig. 4 | Allen frontier-metric sensitivity analyses

a, Consistency between empirical mutual information and balanced accuracy. Each point represents a candidate population or session-window configuration. Empirical mutual information covaried closely with balanced accuracy, supporting the correspondence between the primary task-information proxy and decoding performance. **b, Compact-subset size across time.** Number of neurons required to reach 95% of maximum task information (compact k95) across time windows. Earlier and intermediate windows generally required fewer units, whereas k95 increased in later windows, indicating reduced compressibility over processing time. **c, Relationship between compact k95 and best k.** The x-axis shows compact k95 and the y-axis the population size achieving maximal task information. Most observations lie above the identity line, indicating that once near-maximal information is reached, adding further neurons typically yields only limited or unstable additional information. **d, Marginal residual efficiency.** MRE quantifies the marginal information gain associated with the additional spike-count energy required to expand from the compact subset to all units. Values near or below zero indicate that additional population activity often increases spiking cost without commensurate task-information gain.



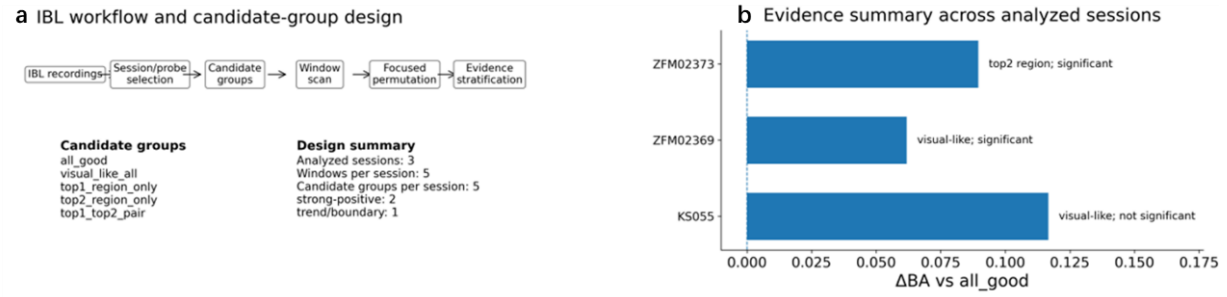
Repository Fig. 5 | Late-window state-stratified frontier envelopes across Allen sessions

Each panel shows the state-stratified energy–information frontier envelope for one Allen session in the late visual-response window (250–400 ms). Blue denotes Stationary trials and red Active trials. The x-axis shows spike-count energy relative to the all-units reference within each state, and the y-axis shows information recovery relative to the maximum task information in that state. Solid lines show the monotonic frontier envelope, light dashed lines the raw top-k energy–information trajectory, and the horizontal dashed line the 95% information-recovery threshold. The degree of Active–Stationary separation varied across sessions: in some recordings, Stationary trials reached 95% information recovery at lower energy ratios, whereas other sessions showed weaker or differently directed effects. These data support the main-text conclusion that late visual frontier geometry is behaviorally modulated but session dependent.



Repository Fig. 6 | Temporal dynamics of state-stratified frontier metrics

Temporal dynamics of multiple frontier metrics for Stationary and Active trials in the Allen dataset. **Top left,** FCI_{95}. **Top right,** compact energy ratio. **Bottom left,** compact k95, defined as the number of neurons required to recover 95% of maximum task information. **Bottom right,** distributed residual index (DRI). Thick lines indicate the cross-session median, thin lines show individual-session trajectories, and shaded regions indicate the interquartile range. Early and intermediate visual-response windows remained characterized by high FCI_{95}, low compact energy ratio, and relatively low k95 in both behavioral states. In the latest window, Active trials showed lower FCI_{95}, increased compact energy ratio, and larger k95. DRI remained near zero in most conditions, indicating that late state modulation primarily reflected slower frontier saturation and greater compact-recovery cost rather than a widespread increase in information outside the compact frontier.

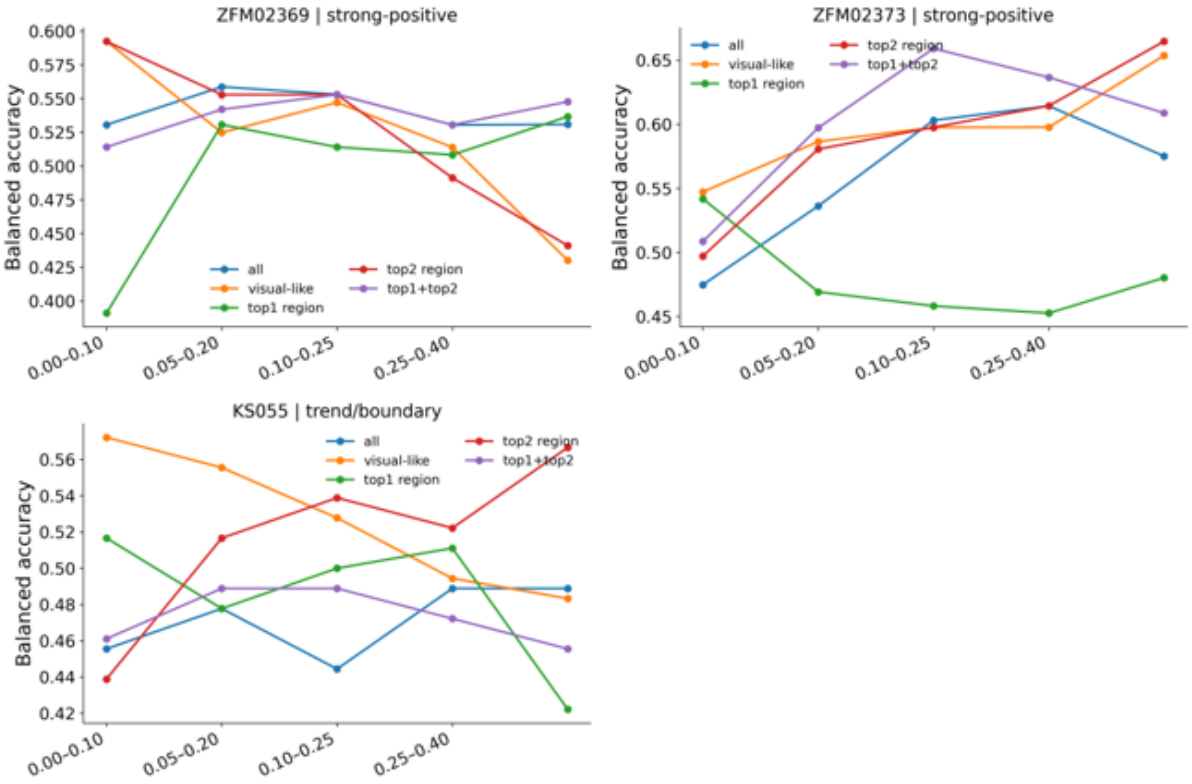


c Session evidence table

session	evidence class	best window	best structured group	ΔBA vs all_good	focused permutation	interpretation
ZFM02373	strong-positive	0.10–0.25	top2_region_only	0.090	best structured significant (p_mi=0.00332); all_good significant (p_mi=0.0266)	strong support
ZFM02369	strong-positive	0.00–0.10	visual_like_all	0.062	best structured significant (p_mi=0.036); all_good not significant (p_mi=0.603)	strong support
KS055	trend/boundary	0.00–0.10	visual_like_all	0.117	best structured not significant (p_mi=0.127); all_good not significant (p_mi=0.349)	positive but boundary

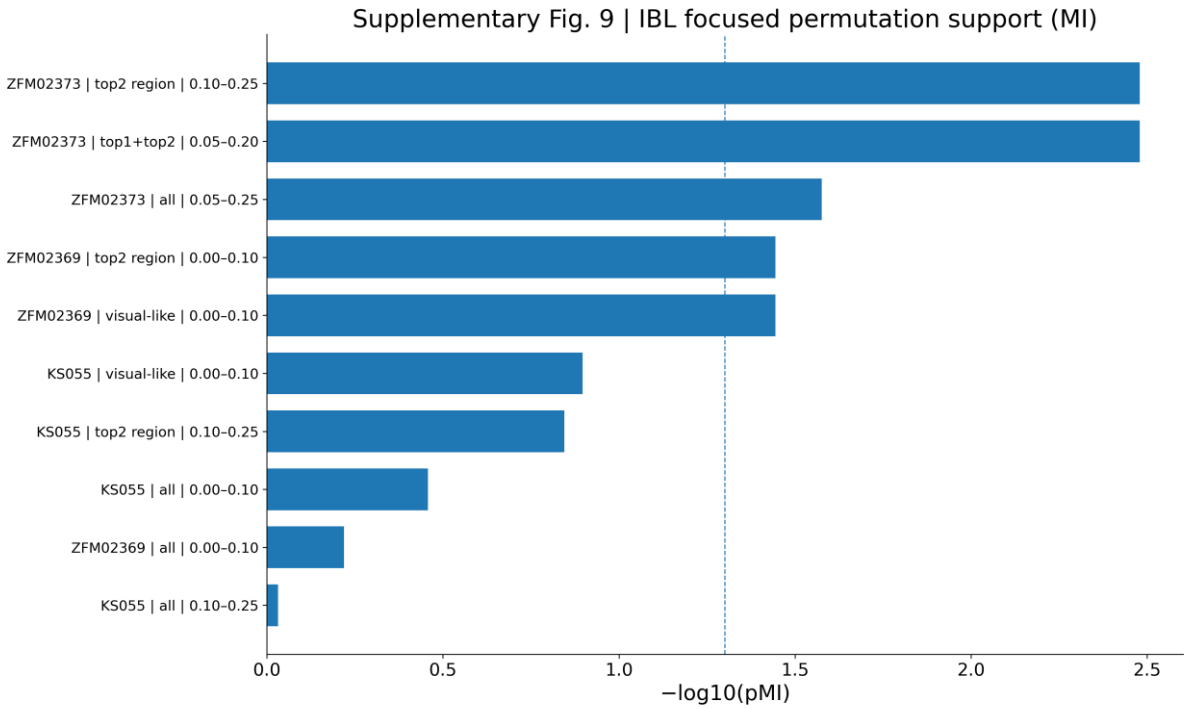
Repository Fig. 7 | IBL analyzed sessions and evidence stratification

a, IBL workflow and candidate-population design. The workflow summarizes recording screening, session/probe selection, candidate-population construction, window scanning, focused permutation testing, and evidence stratification. Candidate populations included all_good, visual_like_all, top1_region_only, top2_region_only, and top1_top2_pair. Three analyzed evidence sessions were retained, each evaluated across five candidate time windows and five candidate neural populations; two sessions were classified as strong-positive and one as trend/boundary. **b, Evidence summary across analyzed sessions.** The x-axis shows the balanced-accuracy gain of the best structured candidate population relative to all_good. ZFM02373 and ZFM02369 received focused-permutation support and were therefore classified as strong-positive, whereas KS055 showed a positive decoding advantage without stable permutation significance and was classified as trend/boundary. **c, Session evidence table.** The table summarizes evidence class, best time window, best structured population, ΔBA relative to all_good, focused-permutation outcome, and interpretation for each analyzed session.



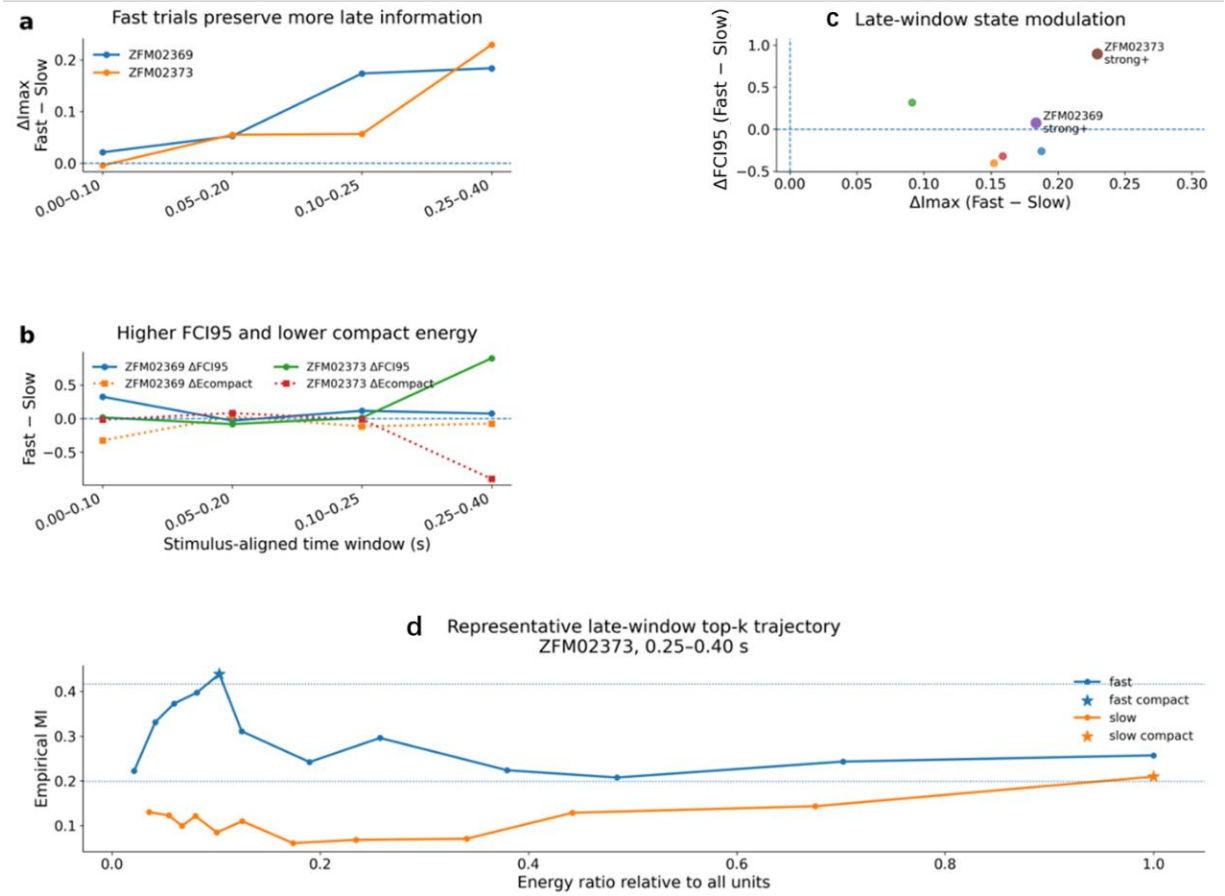
Repository Fig. 8 | Complete IBL window scans across candidate populations

Balanced accuracy of the candidate neural populations across stimulus-response windows in ZFM02369, ZFM02373, and KS055. ZFM02369 showed an early structured advantage at 0.00–0.10 s, particularly for visual_like_all and top2_region_only. ZFM02373 showed stronger structured performance in intermediate windows, including top2_region_only and top1_top2_pair. KS055 showed positive structured trends in some windows but greater variability across candidate populations and time windows. These results demonstrate that expression of the structured visual frontier in IBL was both time-window and session dependent.



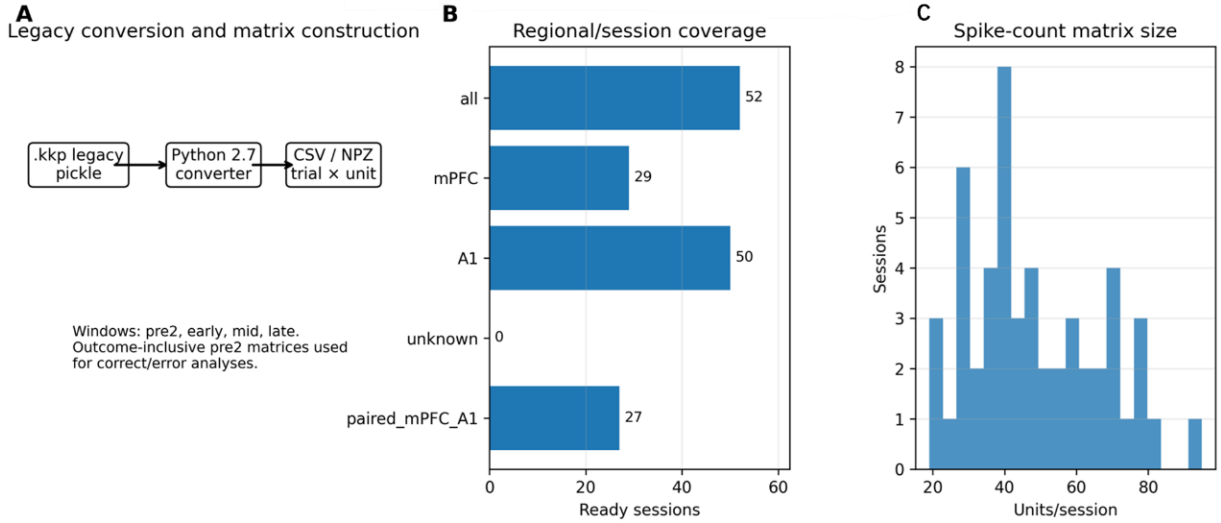
Repository Fig. 9 | IBL focused-permutation support based on empirical mutual information

Focused-permutation results for key IBL candidate configurations. The x-axis shows −log10(pMI), and the dashed line denotes the pMI = 0.05 significance threshold. ZFM02373 top2_region_only and top1_top2_pair exceeded the threshold in the key window. In ZFM02369, visual_like_all and top2_region_only also exceeded the threshold in the 0.00 – 0.10 s window, whereas all_good did not receive significant support. KS055 visual_like_all and top2_region_only showed positive effects but did not cross the significance threshold, supporting its classification as trend/boundary.



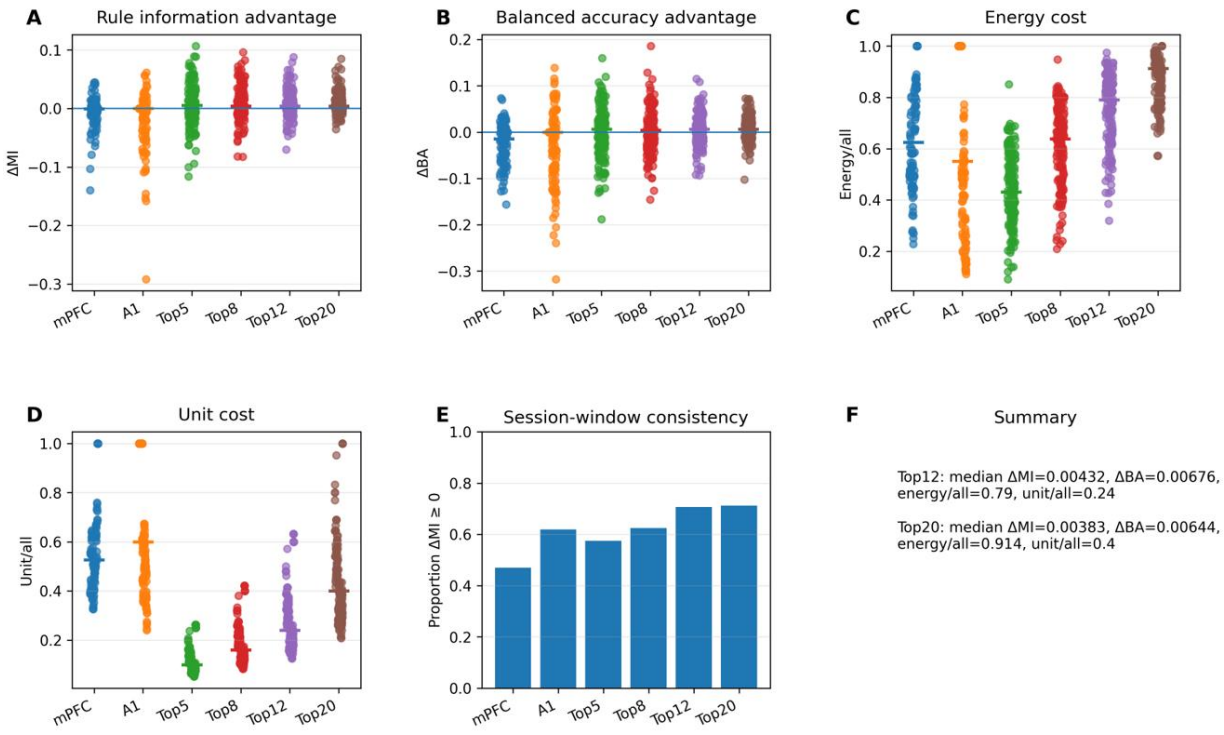
Repository Fig. 10 | IBL Fast/Slow cognitive-state frontier modulation

Supplementary cognitive-state analysis of Fast and Slow trials in the IBL strong-positive sessions. **a,** Fast trials retained more maximal task information than Slow trials in later windows, most clearly in ZFM02373 and ZFM02369. **b,** Fast-minus-Slow differences in FCI_{95} and compact energy ratio indicate that Fast trials showed greater frontier compressibility or lower compact-recovery cost in some strong-positive session-windows. **c,** Cross-session late-window state modulation was heterogeneous for both information and FCI_{95}. **d,** A representative late-window top-k trajectory from ZFM02373 shows greater empirical mutual information in Fast trials at relatively low energy ratios, whereas task information was lower overall in Slow trials. Because these state effects were session and variable dependent in IBL, they are presented as supplementary evidence rather than as a core component of the Fig. 4 conclusion.



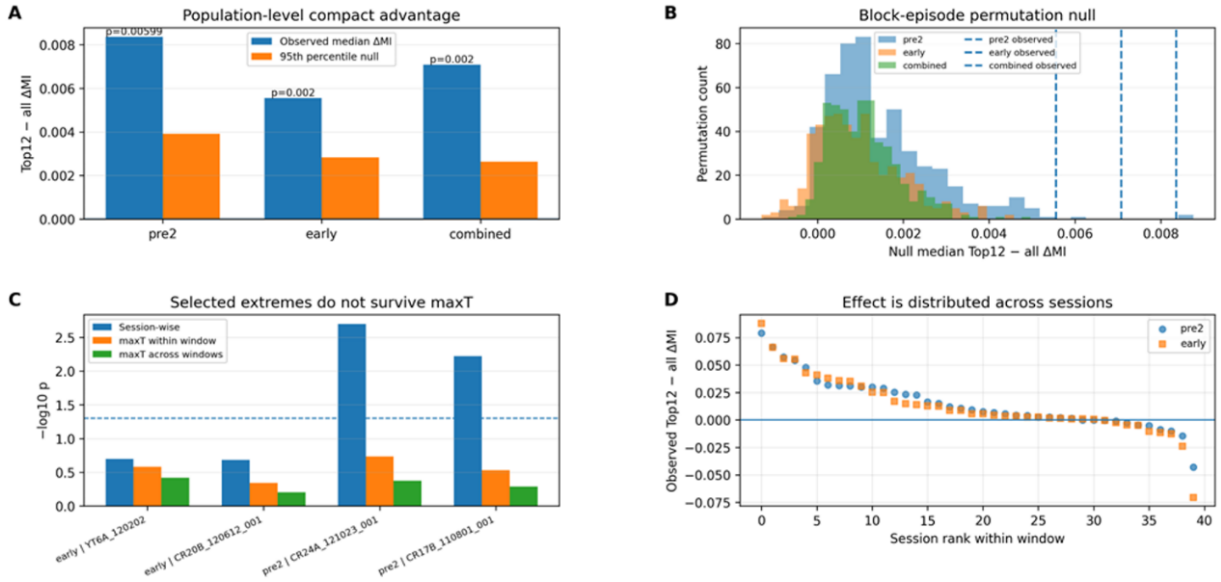
Repository Fig. 11 | CRCNS data conversion and analysis inventory

a, Legacy-data conversion and spike-count-matrix construction. Original CRCNS pfc-1 spike data stored in legacy .kkp pickle files were converted using a Python 2.7 converter into CSV/NPZ formats readable in a modern Python environment and subsequently organized into trial × unit spike-count matrices. The pre2, early, mid, and late windows were used for rule decoding, whereas outcome-inclusive pre2 matrices were used exclusively for correct/error analyses. **b, Regional and session coverage.** Bars show the number of analysis-ready sessions under different regional conditions. Fifty-two sessions were available for all-unit analyses, including 29 mPFC-ready sessions, 50 A1-ready sessions, 27 paired mPFC/A1 sessions, and no sessions with unknown regional labels. **c, Distribution of spike-count-matrix size.** Histogram of the number of analyzable units per trial × unit matrix across available sessions. Most sessions contained tens of units, providing sufficient population size for compact top-k, all-units, and regional analyses.



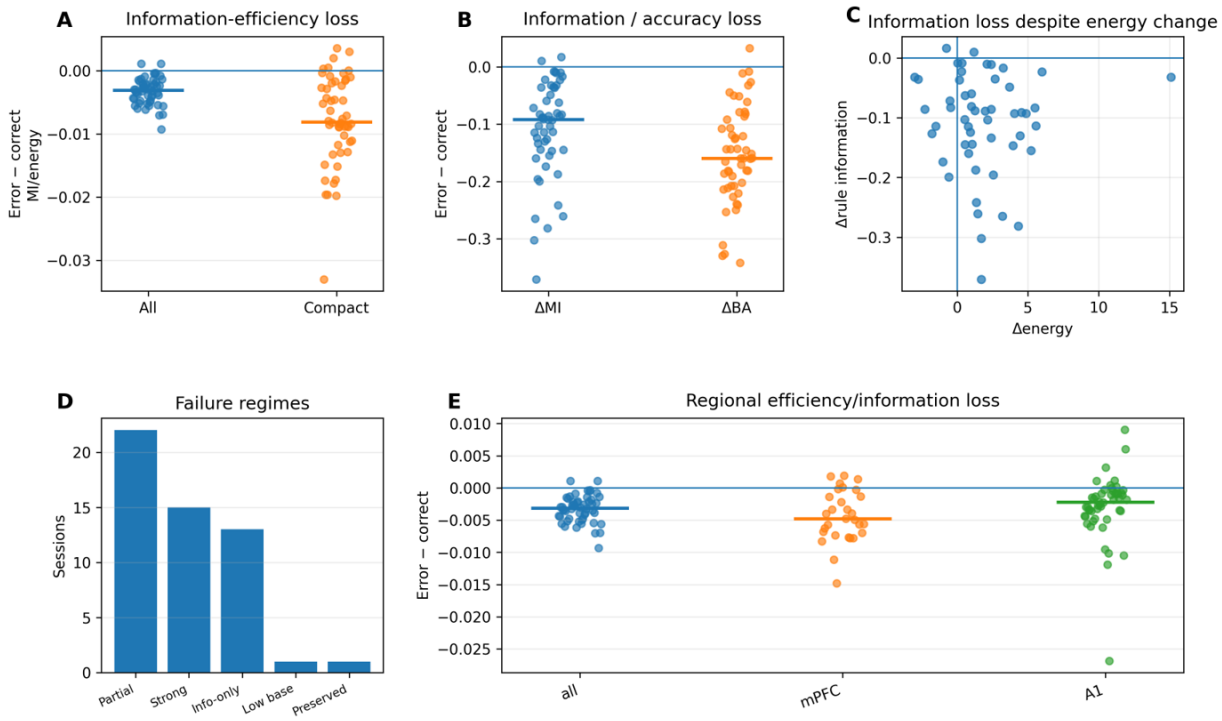
Repository Fig. 12 | Complete CRCNS block-aware cross-validation results

a, Rule-information advantage. Distribution of empirical-MI differences (ΔMI) relative to all_units for fixed candidate populations including mPFC, A1, Top5, Top8, Top12, and Top20. Each point represents one session-window and short horizontal bars denote medians. Several compact top-k populations showed positive or near-zero ΔMI overall, with Top12 and Top20 serving as the principal confirmatory compact subsets in the main text. **b, Balanced-accuracy advantage.** Distribution of balanced-accuracy differences (ΔBA) relative to all_units. Top12 and Top20 showed overall positive advantages, further supporting them as robust compact rule-frontier representatives. **c, Spiking cost.** Spike-count energy relative to all_units (energy/all). Top12 and Top20 both remained below one, indicating that their rule-information performance was not achieved through greater population spiking cost. **d, Unit cost.** Fraction of units relative to all_units (unit/all). Top12 and Top20 used only a subset of the full recorded population. **e, Session-window consistency.** Fraction of session-windows with ΔMI ≥ 0 for each fixed candidate population. Top12 and Top20 matched or exceeded all_units in the majority of session-windows. **f, Summary of Top12 and Top20.** Median values for Top12 were ΔMI = 0.00432, ΔBA = 0.00676, energy/all = 0.79, and unit/all = 0.24; corresponding Top20 values were ΔMI = 0.00383, ΔBA = 0.00644, energy/all = 0.914, and unit/all = 0.40. These results are consistent with Fig. 5 and show that predefined compact Top12/Top20 populations recover rule information using fewer units without higher spike-count cost than all_units.



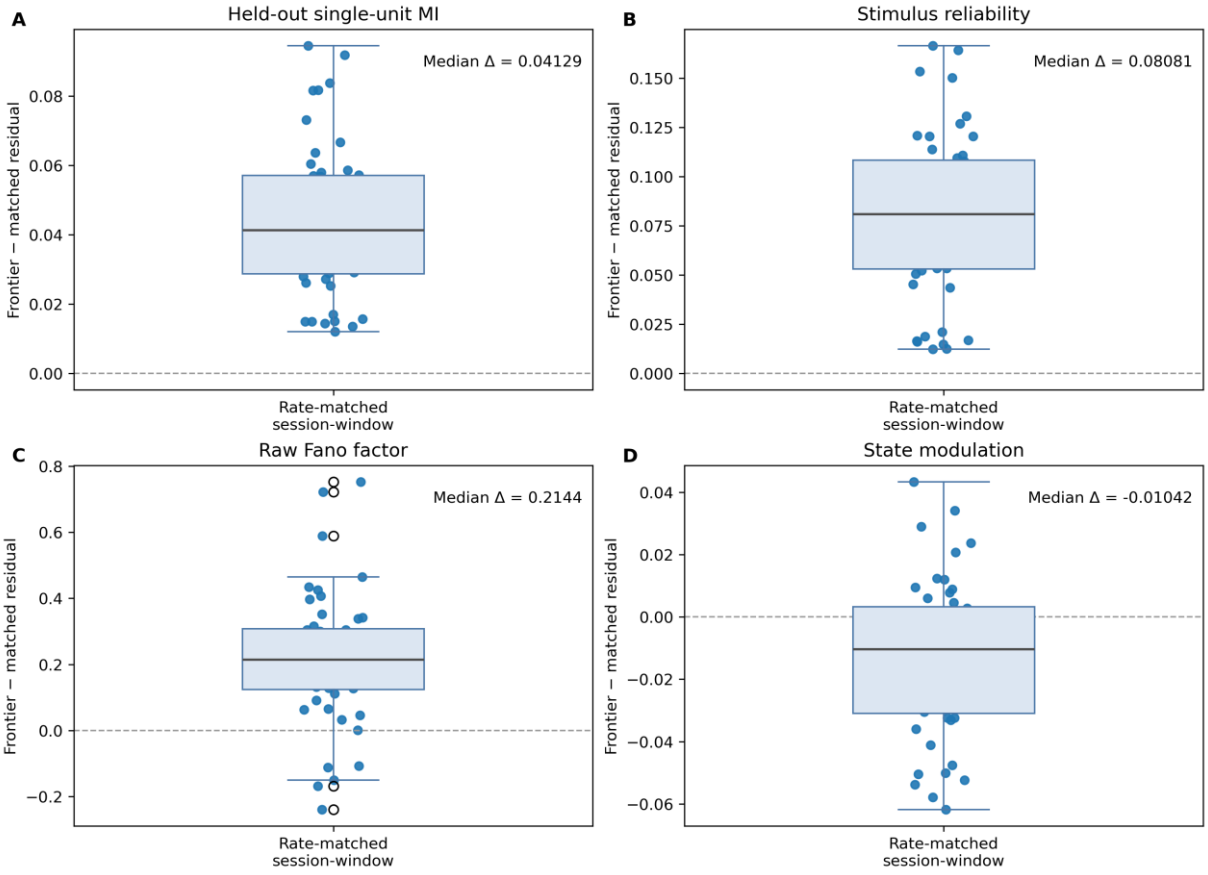
Repository Fig. 13 | CRCNS population-level block-episode permutation and selection-adjusted boundary

a, Population-level compact advantage. Observed median Top12-minus-all_units ΔMI across all 40 block-CV-eligible sessions for the pre2 and early windows and for both windows combined. Orange bars indicate the 95th percentile of the corresponding population-median null distributions. Observed median ΔMI was 0.00835 in pre2 (29/40 sessions positive, p = 0.00599), 0.00555 in early (31/40 positive, p = 0.001996), and 0.00708 across the combined 80 session-windows (60/80 positive, p = 0.001996). **b, Block-episode permutation null distributions.** Null distributions of the population median ΔMI from 500 valid block-episode label permutations for pre2, early, and the combined analysis; dashed lines denote the observed statistics. Each permutation reconstructed the block-aware folds and repeated Top12 neuron ranking using training groups only. **c, Selected extremes do not survive max-statistic correction.** Uncorrected session-wise p values, within-window 40-session max-statistic-corrected p values, and joint max-statistic-corrected p values across the 80 pre2/early candidates are shown for four previously selected extreme session-windows. None survived either max-statistic correction. **d, Effect distribution across sessions.** Session-wise Top12-minus-all_units ΔMI ranked within the pre2 and early windows, showing that the positive population effect was distributed across multiple recordings rather than being independently driven by a few selected extremes. Empirical p values were calculated as (1 + k)/(1 + N); null statistics differing from the observed statistic by no more than 10^{-12} were conservatively treated as ties.



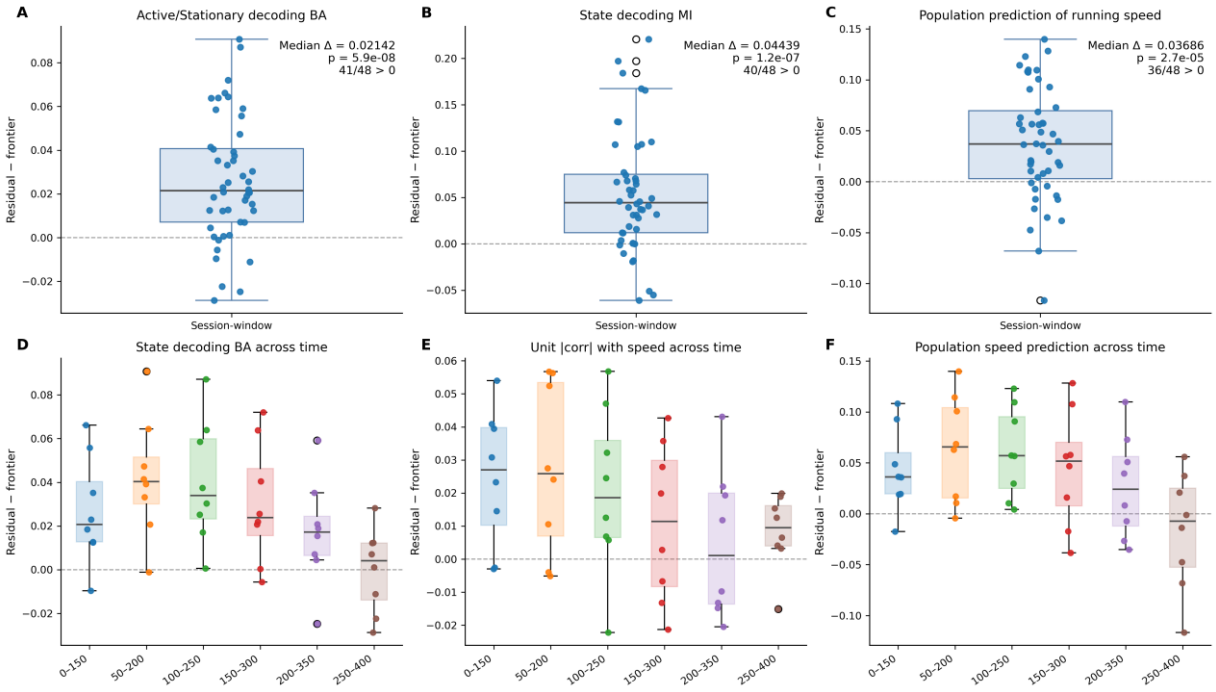
Repository Fig. 14 | CRCNS correct/error information efficiency and failure regimes

a, Information-efficiency loss. Error-minus-correct differences in MI per unit spike-count energy. Both all_units and compact subsets showed predominantly negative values, indicating reduced conversion of spiking activity into rule information during error trials. **b, Loss of information and decoding accuracy.** Error-minus-correct ΔMI and ΔBA were predominantly negative, showing that behavioral failure was accompanied by reduced rule information and rule-decoding accuracy. **c, Information loss despite heterogeneous energy change.** The x-axis shows Δenergy and the y-axis Δrule information. Many sessions showed reduced rule information despite unchanged or increased energy, indicating that errors could not be explained simply by reduced spiking activity. **d, Distribution of failure regimes.** Sessions were classified as partial, strong, information-only, low-baseline, or preserved/compensated. Most sessions fell into partial weakening, strong collapse, or information-only degradation categories, whereas low-baseline and preserved/compensated cases were uncommon. **e, Regional efficiency/information loss.** Error-minus-correct changes were observed in all-unit, mPFC, and A1 analyses, with mPFC showing comparatively stronger negative changes, providing supplementary evidence for regional sensitivity during rule-task failure.



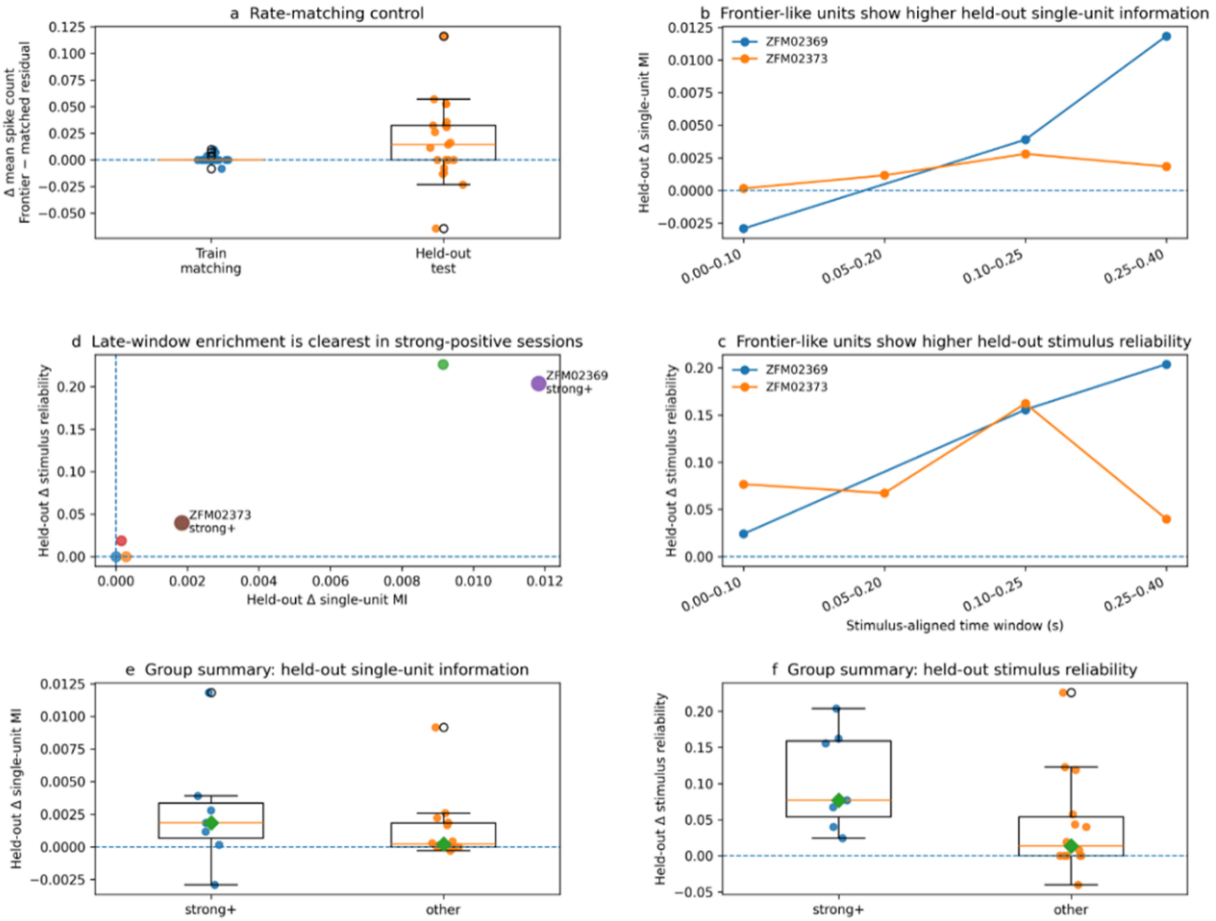
Repository Fig. 15 | Allen frontier-unit functional enrichment and rate-matching diagnostics

a, Held-out single-unit mutual information. Frontier units were compared with mean-spike-count-matched residual units using held-out test folds. Frontier-minus-matched-residual differences were consistently positive, indicating enrichment of visual task information in frontier units. **b, Stimulus reliability.** Frontier units showed greater stimulus reliability than rate-matched residual units, indicating more stable visual responses. **c, Raw Fano factor.** Frontier and matched residual units also differed in raw Fano factor, potentially reflecting differences in stimulus-driven dynamic range; this analysis is treated as a diagnostic rather than a primary conclusion. **d, State modulation.** Frontier-minus-matched-residual differences in state modulation were predominantly negative, indicating stronger behavioral-state modulation in matched residual units. All comparisons were performed within rate-matched session-windows and evaluated on held-out data.



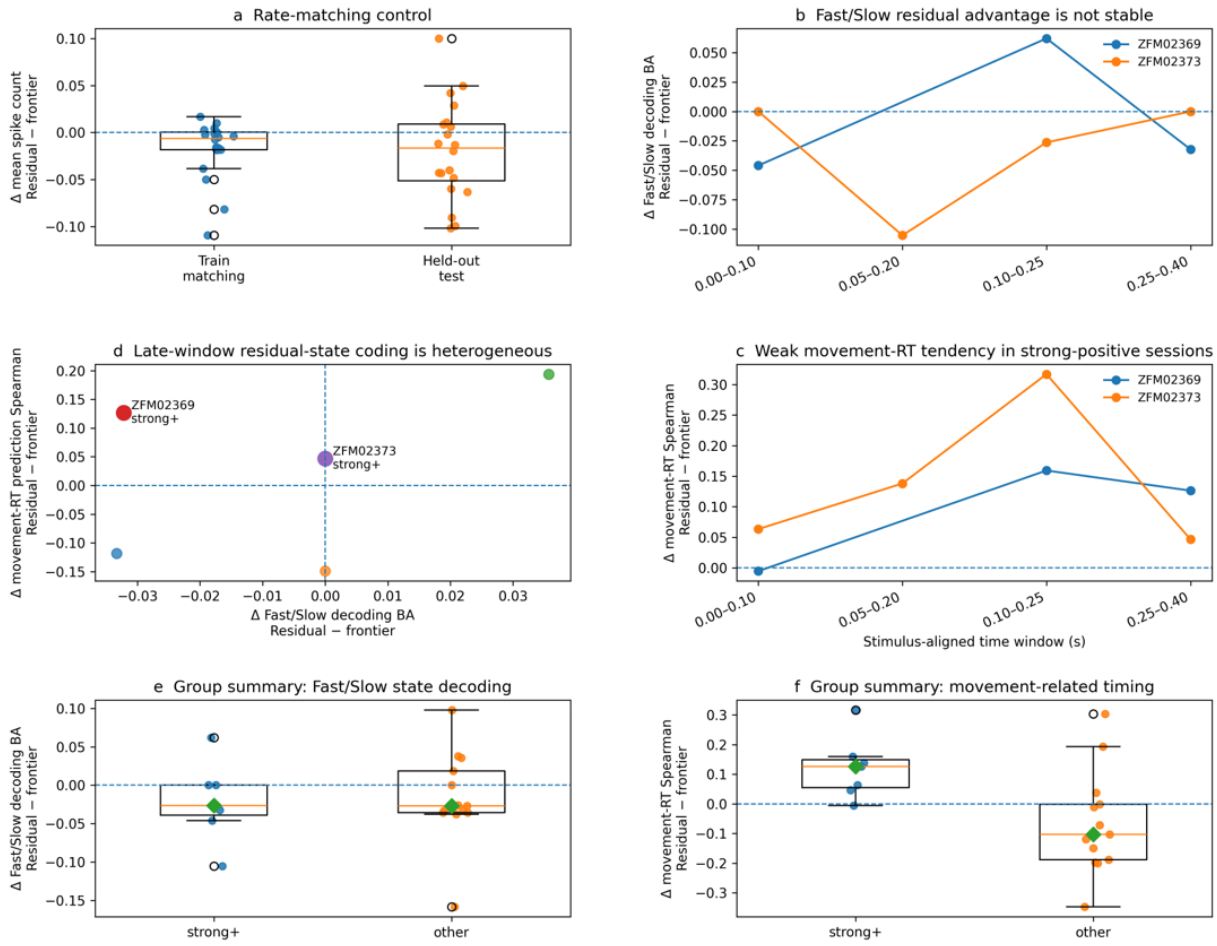
Repository Fig. 16 | Behavioral-state coding in Allen residual units

a, Active/Stationary decoding balanced accuracy. Matched residual populations showed an overall advantage over frontier populations in behavioral-state decoding. **b, State-decoding mutual information.** Residual populations carried greater information about Active/Stationary state. **c, Population prediction of running speed.** Residual populations predicted running speed more accurately than frontier populations, indicating that residual activity was not simply uninformative noise. **d–f, Time-window dependence.** Residual-minus-frontier differences in state-decoding balanced accuracy, unit-level speed correlation, and population-level speed prediction varied across time windows, with residual behavioral-state coding weakening in the latest window. These analyses extend the result shown in Fig. 6d.



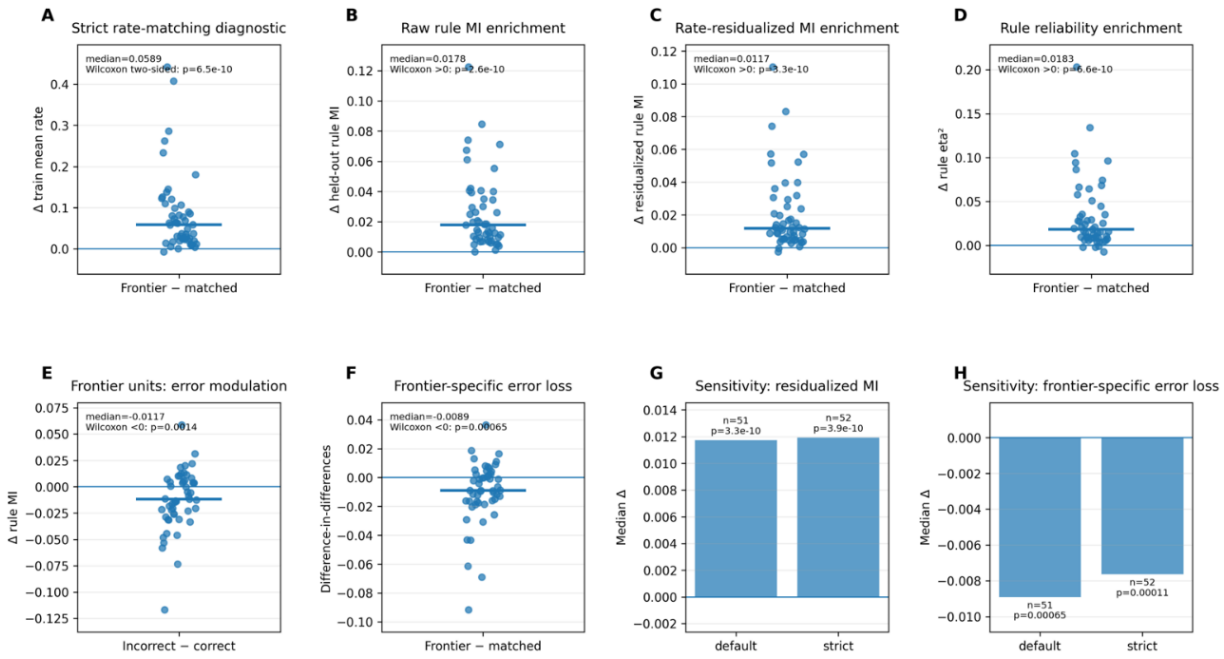
Repository Fig. 17 | Cross-validation-safe IBL validation of frontier-like functional enrichment

a, Rate-matching control. Frontier-like units were defined using training folds, and matched residual units were selected according to training-fold mean spike count; held-out test folds were reserved for functional evaluation. **b, Held-out single-unit information.** Frontier-like units showed greater held-out single-unit stimulus information in IBL strong-positive sessions. **c, Held-out stimulus-response separation.** Frontier-like units showed stronger separation of responses to the two stimulus classes in a subset of session-windows. **d, Late-window enrichment.** Frontier-like functional enrichment was clearest in later windows of the strong-positive sessions, whereas effects were more heterogeneous elsewhere. **e–f, Group summaries.** Strong-positive sessions showed more consistent frontier-like enrichment in held-out single-unit information and stimulus-response separation than other sessions. These results support the use of IBL as an external functional validation dataset while also defining its boundary conditions.



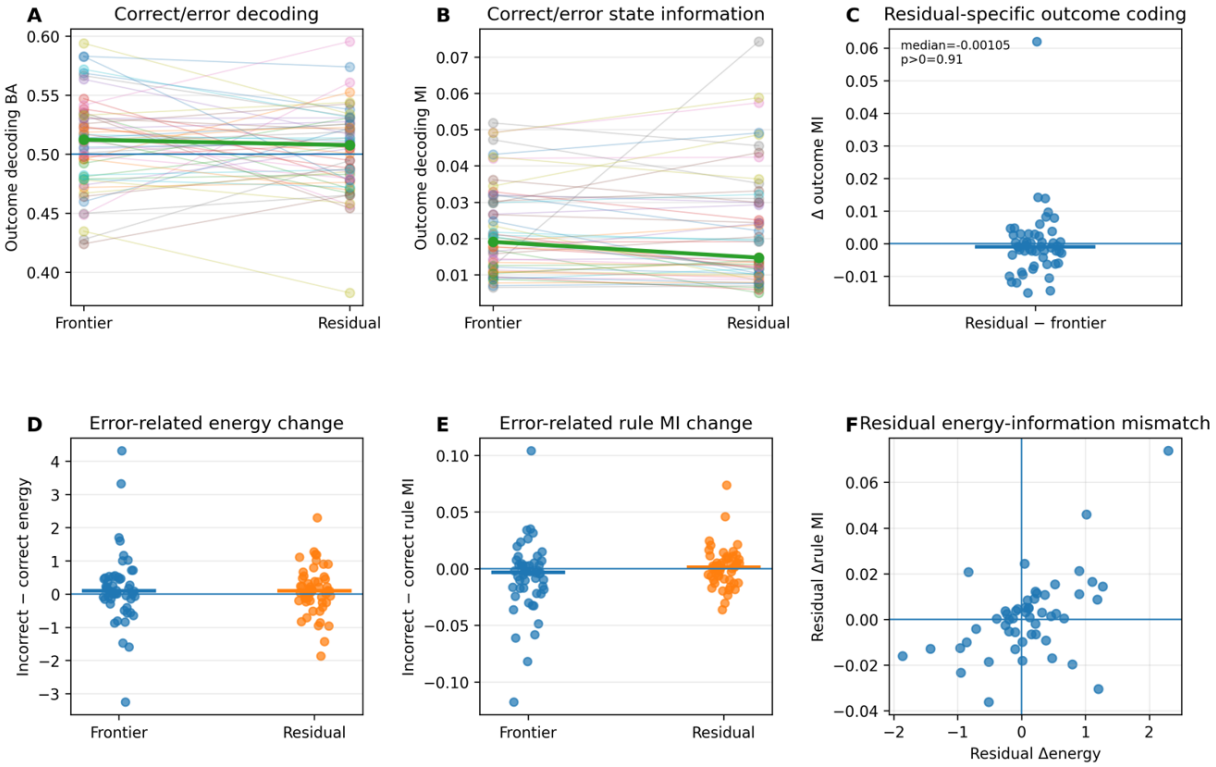
Repository Fig. 18 | Heterogeneous residual behavioral-state coding in IBL

a, Rate-matching control. Residual-like and frontier-like populations were matched by training-fold mean spike count, and behavioral-state metrics were evaluated on held-out folds. **b, Fast/Slow residual advantage.** Residual-minus-frontier differences in Fast/Slow decoding balanced accuracy were inconsistent across windows and sessions. **c, Movement reaction-time tendency.** Strong-positive sessions showed a tendency toward residual advantage in movement-related reaction-time measures, but this effect did not replicate consistently across sessions. **d, Late-window residual-state heterogeneity.** Residual advantages in late-window Fast/Slow decoding and movement-RT prediction varied markedly between sessions. **e–f, Group summaries.** Both Fast/Slow state decoding and movement-related timing measures showed variable effects across strong-positive and other sessions. Thus, residual behavioral-state coding in IBL should be interpreted as dataset- and variable-dependent rather than as a universal mechanism.



Repository Fig. 19 | Functional enrichment and error-related degradation of CRCNS rule-frontier units

a, Strict rate-matching diagnostic. Mean firing rates of rule-frontier and matched residual units were evaluated under strict matching criteria, together with subsequent rate-residualized analyses to control for firing-rate confounding. **b, Raw rule-MI enrichment.** Rule-frontier units showed greater held-out rule mutual information than matched residual units. **c, Rate-residualized MI enrichment.** Rule-frontier units retained greater rule information after controlling for firing rate by residualization. **d, Rule-reliability enrichment.** Rule-frontier units showed greater rule reliability than matched residual units. **e, Error modulation of frontier units.** Rule information in frontier units decreased during incorrect trials. **f, Frontier-specific error loss.** Error-related rule-information loss was greater in rule-frontier than in matched residual units. **g–h, Sensitivity analyses.** Both rate-residualized rule-MI enrichment and frontier-specific error loss were preserved under the default and stricter matching conditions, supporting the robustness of rule-frontier functional enrichment.



Repository Fig. 20 | CRCNS residual units and correct/error state coding

a, Correct/error decoding. Frontier and matched residual populations did not show a clear residual advantage in correct/error decoding balanced accuracy. **b, Correct/error state information.** Residual units did not show greater information about behavioral outcome. **c, Residual-specific outcome coding.** Residual-minus-frontier differences in outcome mutual information were centered near zero, indicating no selective enhancement of correct/error coding in residual units. **d, Error-related energy change.** Frontier and residual populations showed heterogeneous incorrect-minus-correct changes in spike-count energy. **e, Error-related rule-information change.** Residual units did not show a compensatory increase in rule information during incorrect trials. **f, Residual energy–information mismatch.** Changes in residual-population energy were not consistently associated with compensatory changes in residual rule information. These findings support the main-text interpretation that behavioral failure in CRCNS is more consistent with degradation of the rule-informative frontier than with compensatory rule coding by residual populations.

Repository Table 1 | Cross-validation and permutation-test parameters

Dataset/ analysis	Cross-validation design	Feature or subset selection	Permutation test	Exclusion criteria
Allen top-k frontier	Repeated stratified 5-fold CV, 5 repeats, 25 folds in total	Neurons ranked within each training fold by static-grating orientation information	None	Excluded when the number of trials in a behavioral state or stimulus class was insufficient
Allen Active/Stationary	Repeated stratified 5-fold CV within each state, 5 repeats	Neurons re-ranked within each training fold	None	Excluded when state-stratified data did not permit the predefined CV procedure
Allen regional analysis	StratifiedKFold, 5 folds, shuffle=True	Fixed top-12 neurons selected within each candidate pool	30 label permutations for region-pair gain[1]	If a candidate pool contained fewer than 12 but at least 8 valid units, all available units were used
IBL ZFM02373	StratifiedKFold, 5 folds, shuffle=True	Predefined structured candidate populations	300 focused permutations	Excluded when the number of valid trials was insufficient
IBL ZFM02369	StratifiedKFold, 5 folds, shuffle=True	Predefined structured candidate populations	1,000 focused permutations	Same as above
IBL KS055	StratifiedKFold, 5 folds, shuffle=True	Predefined structured candidate populations	1,000 focused permutations	Same as above
CRCNS primary rule frontier	StratifiedGroupKFold, adaptive 2–5 folds	Top-5/8/12/20 populations constructed within training groups	500 block-episode permutations for representative configurations	Excluded when fewer than two valid folds could be formed
CRCNS correct/error	Stratified CV, up to 5 folds, 5 repeats	Candidate subsets constructed within each training fold	None	Each outcome condition required ≥30 trials and each rule class ≥5 trials

Note: [1] Because each permutation required re-estimation of the best single-region and best region-pair configurations within the fold-wise feature-selection and cross-validation framework, 30 permutations were used for this analysis. The resulting permutation p values were interpreted as descriptive support rather than high-resolution inferential tests.

Repository Table 2 | Classifier and software implementation parameters

Dataset / analysis	Classifier pipeline	Solver	Class weight	Maximum iterations	Random seed
Allen top-k and state analyses	StandardScaler + LogisticRegression	lbfgs	balanced	2000	20260531
Allen regional analysis	StandardScaler + LogisticRegression	lbfgs	None	3000	42
Allen region-pair permutation	Same as regional analysis	lbfgs	None	3000	42
IBL ZFM02373	StandardScaler + LogisticRegression	lbfgs	None	2000	42
IBL ZFM02369	StandardScaler + LogisticRegression	liblinear	None	2000	42
IBL KS055	StandardScaler + LogisticRegression	liblinear	None	2000	42
CRCNS primary rule frontier	StandardScaler + LogisticRegression	liblinear	balanced	1000	0
CRCNS correct/error	StandardScaler + LogisticRegression	liblinear	balanced	1000	20260608

Repository Table 3 Frontier–residual definitions and rate-matching parameters

Dataset / analysis	Matching variable	Matching procedure	Replacement	Repeats	Quality control
Allen stimulus information and reliability	mean_count_cv	Within each session-window, units were divided into up to 10 quantile bins, and residual controls were sampled within each bin	Allowed within bins when insufficient residual units were available	300 repeats	Units with intermediate frontier-selection frequency were excluded from the comparison
Allen state coding	mean_count_cv	Same as above	Allowed within bins when insufficient residual units were available	50 repeats	Same as above
IBL functional validation	Training mean spike count	One-to-one nearest-neighbor matching	Not allowed	At least 25 matched pairs	Only session-windows with ≥25 matched pairs and an absolute difference in median training mean spike count ≤0.15 between frontier-like and matched residual units were included
CRCNS functional validation	Training mean firing rate	Matching caliper defined as the larger of 0.10 and 0.25 × the standard deviation of training mean firing rate across all valid units	Not allowed	At least 4 matched pairs per split	Frontier units without an eligible residual match within the caliper were excluded

Repository Table 4 Definitions of held-out functional metrics

Dataset	Metric	Definition	Interpretation
Allen	Stimulus reliability index	$\mathrm{Var}\mathbb{E}[r \mid s]/\mathrm{Var}(r)$	Fraction of single-unit response variance explained by static-grating orientation
IBL	Binary stimulus-response separation	$\lvert\bar{x}_1 - \bar{x}_0\rvert/(0.5(s_0 + s_1) + 10^{-8})$ [1]	Absolute standardized separation of single-neuron responses between the two stimulus classes
CRCNS	Rule reliability	$\eta^2 = SS_{\text{between}}/SS_{\text{total}}$ [2]	Fraction of single-unit response variance explained by rule identity

[1] $\bar{x}_0$ and $\bar{x}_1$ denote the mean single-unit responses for the two stimulus classes, and s_0 and s_1 denote the corresponding response standard deviations.

[2] $SS_{\text{between}} = \sum_c n_c\,(\bar{x}_c - \bar{x})^2, SS_{\text{total}} = \sum_i (\,x_i - \bar{x})^2$